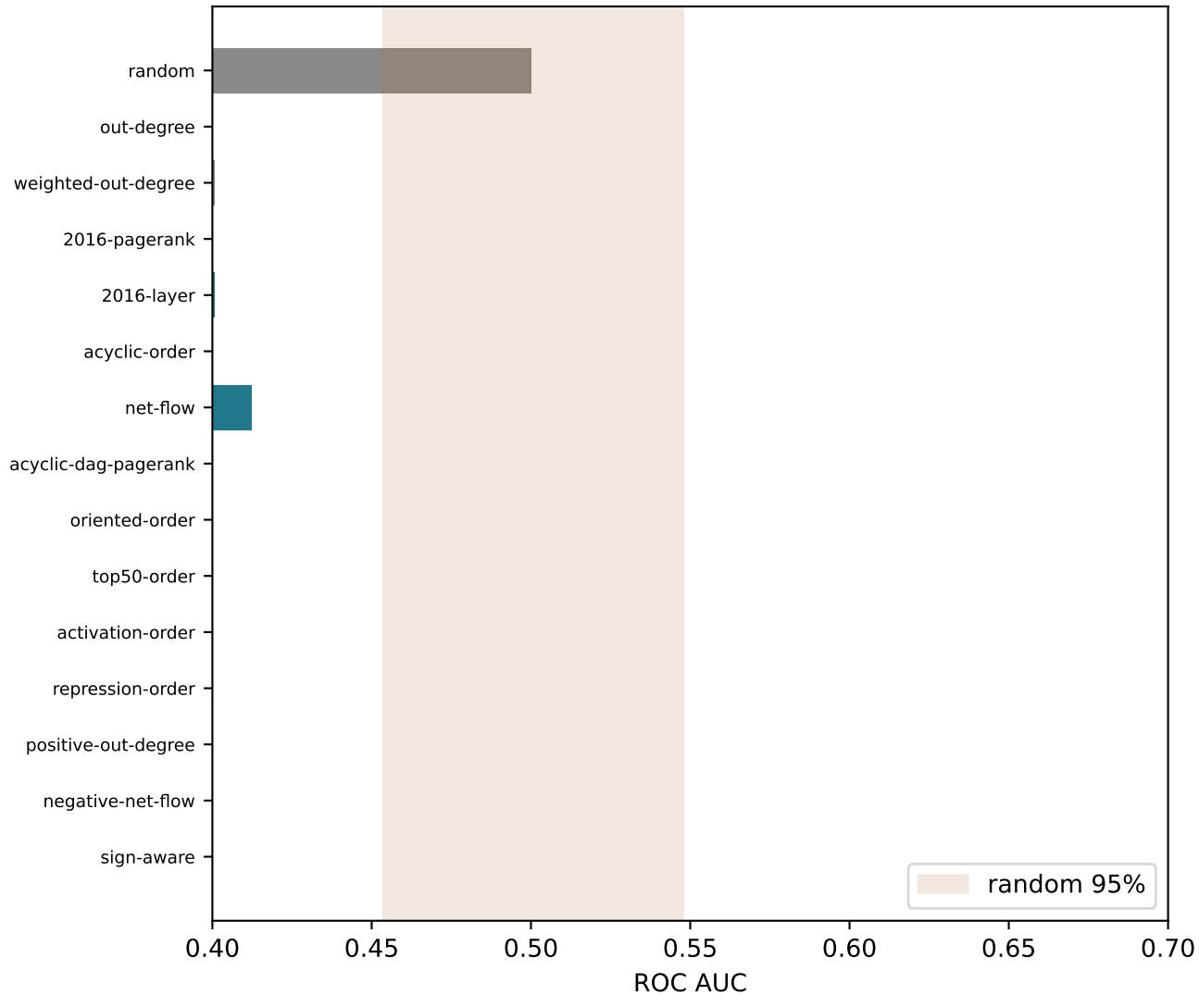
Regulator ranking against reference 'regulators'

2026-11-07 · 739 genes ranked, 199 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	26.8 (19–35)	0.500 (0.454–0.548)				
out-degree	3	0.300	1.000	1.000	-0.100 (-0.119, -0.081)	< 0.001
weighted-out-degree	7	0.401	1.000	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	16	0.367	0.999	1.000	-0.033 (-0.077, +0.007)	0.098
2016-layer	65	0.401	< 0.001	1.000	-0.000 (-0.046, +0.041)	0.984
acyclic-order	9	0.252	1.000	1.000	-0.149 (-0.185, -0.118)	< 0.001
net-flow	8	0.412	1.000	1.000	+0.012 (-0.001, +0.024)	0.058
acyclic-dag-pagerank	7	0.270	1.000	1.000	-0.131 (-0.156, -0.109)	< 0.001
oriented-order	11	0.275	1.000	1.000	-0.125 (-0.164, -0.089)	< 0.001
top50-order	22	0.295	0.899	1.000	-0.106 (-0.152, -0.063)	< 0.001
activation-order	13	0.271	1.000	1.000	-0.129 (-0.166, -0.095)	< 0.001
repression-order	7	0.315	1.000	1.000	-0.085 (-0.121, -0.054)	< 0.001
positive-out-degree †	4	0.365	1.000	1.000	-0.035 (-0.058, -0.012)	0.004
negative-net-flow †	7	0.356	1.000	1.000	-0.045 (-0.071, -0.020)	< 0.001
sign-aware †	4	0.354	1.000	1.000	-0.047 (-0.061, -0.033)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.

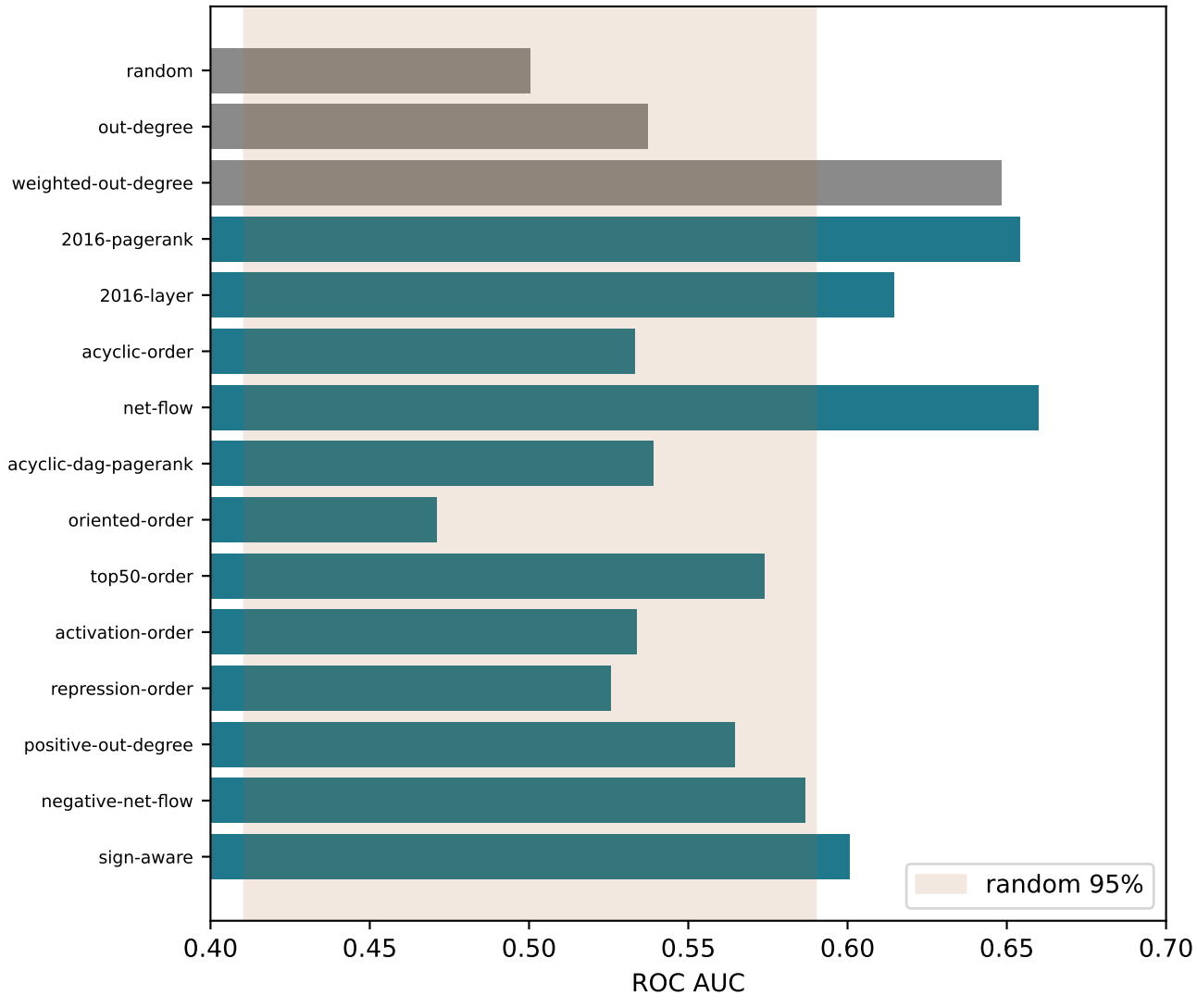


Regulator ranking against reference 'top-layer'

2026-11-07 · 739 genes ranked, 42 in top-regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	5.6 (2–10)	0.500 (0.411–0.590)				
out-degree	2	0.537	0.982	0.206	-0.111 (-0.152, -0.071)	< 0.001
weighted-out-degree	6	0.648	0.507	< 0.001	+0.000 (+0.000, +0.000)	
2016-pagerank	11	0.654	0.019	< 0.001	+0.006 (-0.062, +0.077)	0.906
2016-layer	30	0.615	< 0.001	0.007	-0.034 (-0.111, +0.044)	0.364
acyclic-order	6	0.533	0.507	0.237	-0.115 (-0.177, -0.060)	< 0.001
net-flow	6	0.660	0.507	< 0.001	+0.012 (-0.005, +0.029)	0.188
acyclic-dag-pagerank	4	0.539	0.843	0.194	-0.109 (-0.149, -0.072)	< 0.001
oriented-order	6	0.471	0.507	0.741	-0.177 (-0.260, -0.098)	< 0.001
top50-order	11	0.574	0.019	0.046	-0.074 (-0.153, +0.005)	0.074
activation-order	7	0.534	0.341	0.233	-0.115 (-0.177, -0.056)	0.002
repression-order	6	0.526	0.507	0.291	-0.123 (-0.185, -0.062)	< 0.001
positive-out-degree †	3	0.565	0.938	0.071	-0.084 (-0.132, -0.040)	< 0.001
negative-net-flow †	5	0.587	0.688	0.029	-0.062 (-0.096, -0.029)	< 0.001
sign-aware †	3	0.601	0.938	0.015	-0.048 (-0.075, -0.022)	0.002

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

true

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	3,641	0.799 (43.13×)	50.15×	32.79×	1.00 (0.57)
resolve_reciprocals → drop_sinks → prune_triangles	2,392	0.110 (33.73×)	42.62×	24.46×	1.00 (0.54)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	947	0.115 (38.50×)	47.78×	30.16×	1.00 (0.52)
absolute	3,641	0.799 (43.13×)			
absolute → minimum_feedback_order	3,641	0.589 (39.72×)			
orient_reciprocals ratio=2.0 → absolute	3,641	0.520 (39.44×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	3,641	0.520 (39.44×)			
top_outgoing k=50 → absolute	3,641	0.799 (50.60×)			
top_outgoing k=50 → absolute → minimum_feedback_order	3,641	0.589 (47.07×)			
keep_sign sign=1 → absolute	3,641	0.553 (50.15×)			
keep_sign sign=1 → absolute → minimum_feedback_order	3,641	0.414 (47.32×)			
keep_sign sign=-1 → absolute	3,641	0.246 (32.79×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	3,641	0.177 (29.06×)			

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 41,927 edges (19,705 negative), largest strong component 739.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 41,927 edges (19,705 negative), largest strong component 739.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 7,759 edges (4,209 negative), largest strong component 109.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 7,759 edges (4,209 negative), largest strong component 109; DAG keeps 3,633 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 41,927 edges (0 negative), largest strong component 739; DAG keeps 36,897 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 41,927 edges (0 negative), largest strong component 739.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 41,927 edges (0 negative), largest strong component 739; DAG keeps 36,897 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 32,088 edges (0 negative), largest strong component 1; DAG keeps 32,088 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 27,677 edges (0 negative), largest strong component 739; DAG keeps 23,623 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 22,222 edges (0 negative), largest strong component 731; DAG keeps 19,376 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 19,705 edges (0 negative), largest strong component 729; DAG keeps 17,530 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 41,927 edges (19,705 negative), largest strong component 739.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 41,927 edges (19,705 negative), largest strong component 739.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 41,927 edges (19,705 negative), largest strong component 739.